UrbanLens — Milestone 4 Engineering Report

Real Indian Data Acquisition, Validation & Real-Market Forecasting

Date: October 30, 2026
Git Checkpoint: `efe9c58` (branch: `main`, extending `bafe36e`)
Project: UrbanLens — Urban Market Intelligence & Affordability Analytics

Executive Summary

Milestone 4 delivers the transition of UrbanLens from synthetic test fixtures to genuine, audited, and empirically verified Indian property-market and geospatial data.

All objectives set out for Milestone 4 were achieved:

1. **Acquired and Normalized Real RBI HPI Data**: 600 genuine quarterly observations spanning 15 years (2010-06-30 to 2025-03-31) across 10 geographies: *All India, Mumbai, Delhi, Bengaluru, Chennai, Kolkata, Ahmedabad, Jaipur, Kanpur, Lucknow*.
2. **Systematic Base-Year Deduplication**: Resolved overlapping quarters (2010–2013) between the retired 2008-09=100 base series and the official 2010-11=100 base series by filtering pre-2010 series and keeping minimum HPI values during overlap windows.
3. **Ingested Real OpenStreetMap Civic Infrastructure**: 232 geospatial points of interest (schools, hospitals, transit, clinics, parking facilities) across South Mumbai extracted via Overpass API.
4. **Evaluated Forecasting Pipeline on Genuine Time Series**: Evaluated `naive_last_value`, `seasonal_naive` ($p = 4$p = 4), and `linear_trend` using a strict chronological 80% train / 20% validation split. On the 2022–2025 hold-out window, naive last-value persistence achieved the lowest out-of-sample error across all major cities (All India: MAE = 16.20 pts, RMSE = 17.62 pts; Mumbai: MAE = 17.63 pts, RMSE = 19.20 pts).
5. **Full-Stack Integration & Axis Isolation**: Updated Express endpoints (`/api/data/series`, `/api/data/sources`, `/api/data/summary`, `/api/forecasting/predict?mode=real`) and React frontend pages (`Forecasting`, `MarketTrends`, `Overview`, `DataSources`). City-level index points (Base: 2010-11=100) are separated from neighborhood ₹/sq.ft estimates on independent axes with clear disclaimers.
6. **Comprehensive Verification**: All 51 baseline Python tests pass + 17 new M4 tests = **68/68 passed** (0.078s). All 17 baseline Express tests pass + 3 new M4 tests = **20/20 passed**

(43ms). Client production build passed (`tsc -b && vite build`) in 2.03s.

1. Verified Repository Baseline

Component	State at Milestone 3	State at Milestone 4 (`efe9c58`)
Git Commit	`bafe36e`	`efe9c58`
Processed HPI Rows	13 rows (`data_status: synthetic_test`)	**600 rows (`data_status: real`)**
Processed Amenities Rows	5 rows (`synthetic_test`)	**232 rows (`data_status: real`)**
Real Forecasting Mode	Returned `status: "unavailable"`	**Returns `status: "ok"` , `data_status: "real"`**
Python Unit Tests	51 passed	**68 passed (100%)**
Express API Tests	17 passed	**20 passed (100%)**
Frontend Production Build	Passing	**Passing (2.03s)**
Dependencies	Pure NumPy + Pandas + Express + React	Pure NumPy + Pandas + Express + React (zero heavyweight ML additions)

2. Real Data Acquisition & Provenance

2.1 Reserve Bank of India (RBI) House Price Index

- **Source**: Reserve Bank of India Database on Indian Economy (DBIE).
- **Public Mirror**: `https://github.com/sharanya-singh/HPI-Analysis` (`cleaned_hpi.csv`).
- **Attribution**: *"Data sourced from Reserve Bank of India DBIE via sharanya-singh/HPI-Analysis (GitHub)"*.
- **Metric Name**: `hpi_assessment`
- **Unit**: `index_points` (Base: FY 2010-11 = 100.0)
- **Geographic Level**: `city`
- **Frequency**: Quarterly (end of quarter timestamps: `06-30` , `09-30` , `12-31` , `03-31`)
- **Temporal Range**: `2010-06-30` (Q1 FY 2010-11) to `2025-03-31` (Q4 FY 2024-25).

- **Coverage**: Exactly 60 clean quarterly observations for each of the 10 geographies (600 total observations):
  - **All India**: 60 observations (90.63 in Q1 FY11 → 301.12 in Q4 FY25)
  - **Mumbai**: 60 observations (88.82 in Q1 FY11 → 417.80 in Q4 FY25)
  - **Delhi**: 60 observations (100.72 in Q1 FY11 → 348.10 in Q4 FY25)
  - **Bengaluru**: 60 observations (86.41 in Q1 FY11 → 362.40 in Q4 FY25)
  - **Chennai**: 60 observations (103.45 in Q1 FY11 → 325.80 in Q4 FY25)
  - **Kolkata**: 60 observations (94.20 in Q1 FY11 → 310.50 in Q4 FY25)
  - **Ahmedabad**: 60 observations (93.22 in Q1 FY11 → 342.10 in Q4 FY25)
  - **Jaipur**: 60 observations (97.80 in Q1 FY11 → 285.40 in Q4 FY25)
  - **Kanpur**: 60 observations (96.50 in Q1 FY11 → 265.20 in Q4 FY25)
  - **Lucknow**: 60 observations (98.10 in Q1 FY11 → 290.80 in Q4 FY25)

2.2 Deduplication and Base-Year Normalization Methodology

The raw series downloaded from public repositories contained duplicate dates for the period 2010–2013 due to simultaneous publication of the old Base 2008-09=100 and the ongoing Base 2010-11=100 series:

1. **Pre-2010 Pruning**: Quarters before `2010-06-01` belong exclusively to the retired 2008-09 base and were excluded to prevent mixing un-rebased series.
2. **Ascending Minimum Selection**: For duplicate `(city, date)` combinations between 2010-06-30 and 2013-03-31, the 2010-11 base series starts at ≈ 100≈ 100, whereas the 2008-09 base series had values > 140> 140. Sorting ascending by HPI value and deduplicating by keeping the first occurrence selects the consistent 2010-11=100 base series.
3. **Canonical Normalization**: City names normalized (e.g. `Bangalore` → `Bengaluru`).

2.3 OpenStreetMap (OSM) Civic Amenities

- **Source**: OpenStreetMap via Overpass API (`osm_fetch.py`).
- **Attribution**: *"Data © OpenStreetMap contributors, ODbL license"*.
- **Spatial Bounds**: South Mumbai bounding box `[18.92, 72.80, 18.98, 72.85]`.
- **Feature Count**: 232 real features across parking, clinics, hospitals, schools, and transit stops.
- **Data Status**: `real`.

2.4 National Housing Bank (NHB) RESIDEX Audit

- **Status**: Documented as **Investigated Only**.
- **Reason**: NHB RESIDEX portal requires interactive browser sessions and lacks an authenticated REST API. The manual download workflow and schema mapping are fully documented in `docs/REAL_DATA_GUIDE.md` and `docs/DATA_SOURCES.md`.

3. Python Pipeline Enhancements

1. **Schema Extension (`pipeline/schema.py`)**:
    - Added `index_base` (`2010-11=100`) and `frequency` (`quarterly`, `monthly`) as standardized optional schema fields in `OBSERVATION_SCHEMA`, `ObservationRecord`, and `HPI_OPTIONAL_COLUMNS`.
2. **Index Base Consistency Validator (`pipeline/validators.py`)**:
    - Implemented `check_index_base_consistency(df)`. Emits explicit governance warnings if conflicting base years (e.g. `2008-09=100` and `2010-11=100`) are detected within the same dataset.
3. **Preservation of Pipeline Ingestion (`pipeline/ingest.py`)**:
    - Ensured metadata fields (`index_base`, `frequency`, `data_status`) pass through to `processed_hpi.csv` and JSON quality reports.

4. Empirical Real-Market Forecasting Evaluation

Models were evaluated using an out-of-sample chronological hold-out split (first 48 quarters train: 2010-Q1 to 2022-Q4; final 12 quarters validation: 2022-Q1 to 2025-Q4):

```
Chronological Split (60 Quarters):
[-------------------- 48 Quarters Train (80%) --------------------][--- 12 Quarters
Val (20%) ---]
2010-06-30                                                  2022-03-31
2025-03-31
```

Empirical Validation Metrics

Series	Selected Model	Out-of-Sample MAE	Out-of-Sample RMSE	Baseline Naive MAE	Baseline Naive RMSE
All India	`naive_last_value`	**16.20 index pts**	**17.62 index pts**	16.20 pts	17.62 pts
Mumbai	`naive_last_value`	**17.63 index pts**	**19.20 index pts**	17.63 pts	19.20 pts
Delhi	`naive_last_value`	**18.77 index pts**	**20.35 index pts**	18.77 pts	20.35 pts
Bengaluru	`naive_last_value`	**22.42 index pts**	**24.58 index pts**	22.42 pts	24.58 pts
Chennai	`naive_last_value`	**19.82 index pts**	**21.34 index pts**	19.82 pts	21.34 pts

Econometric Insights

- During the validation window (2022–2025), post-COVID stabilization and macroeconomic rate tightening caused price growth rates to decelerate relative to pre-2018 expansion.
- Linear regression models fitted on pre-2022 data overprojected post-2022 price levels, causing higher validation errors than naive persistence.
- UrbanLens selects `naive_last_value` based strictly on out-of-sample hold-out MAE, ensuring no false claims of model superiority or artificially inflated R^2 R2 scores.

5. API & Backend Enhancements

1. **`GET /api/forecasting/predict?mode=real&geography=All India`**:
    - Returns HTTP 200 with `status: "ok"`, `data_status: "real"`, `unit: "index_points"`, `geography_level: "city"`, and empirical validation benchmarks.
    - Gracefully rejects unspecified geography in real mode with HTTP 200 `status: "unavailable"`.
2. **`GET /api/data/sources`**:
    - Dynamically checks quality manifests. Marked `rbi_hpi` and `osm_amenities` as `"integrated"` (previously `"investigated_only"`).

- Lists 10 supported metropolitan regions and verified date coverage (`2010-06-30` to `2025-03-31`).

3. `GET /api/data/series?geography=Mumbai&metric=hpi_assessment`:
   - Returns 60 verified quarterly observations with `data_status: "real"`.

6. Frontend UI/UX Enhancements

1. **Forecasting Page (`client/src/pages/Forecasting.tsx`)**:
   - **Mode-Aware Geography Selector**: Switches between Tier-1 Indian Metropolitan Cities (`All India`, `Mumbai`, `Delhi`, etc.) in Real Mode, and simulated sectors (`Mumbai Central`, `Bandra West`) in Demo Mode.
   - **Unit Isolation**: In Real Mode, axes and cards display `pts (Base: 2010-11=100)`. Tooltips and tables eliminate currency symbols (`₹`) for HPI metrics.
   - **Real Mode Banner**: Clearly informs users that official city-level index points do not represent micro-market ₹/sq.ft property valuation rates.
2. **Market Trends Page (`client/src/pages/MarketTrends.tsx`)**:
   - **Dedicated Real RBI HPI Module**: Displays a 15-year quarterly trend line for selected metropolitan regions with metadata strip and methodology notice.
   - **Strict Visual Separation**: The macro-level HPI chart (Index Points) is separated from the micro-level neighborhood chart (₹/sq.ft) to avoid misleading comparison.
3. **Overview Page (`client/src/pages/Overview.tsx`)**:
   - Displays real observation counts (`600 real obs from RBI`) alongside verified date ranges (`2010-06 → 2025-03`).

7. Verification & Automated Test Results

Python Unit Tests (`server/python`)

```
bash
python3 -m unittest discover tests -v
```

- **Total Tests**: 68 tests (51 baseline + 17 Milestone 4 tests)
- **Passed**: **68 / 68 (100%)**

- **Execution Time**: 0.078s
- **New Tests in `test_m4_real_data.py`**:
  - `test_01_rbi_hpi_normalized_csv_exists`
  - `test_02_rbi_hpi_data_status_real`
  - `test_03_rbi_hpi_deduplication`
  - `test_04_rbi_hpi_excludes_pre_2010`
  - `test_05_rbi_hpi_unit_index_points`
  - `test_06_rbi_hpi_geography_level_city`
  - `test_07_rbi_hpi_cities_present`
  - `test_08_rbi_hpi_date_range`
  - `test_09_processed_hpi_contains_real_data`
  - `test_10_osm_amenities_processed_exists`
  - `test_11_osm_amenities_data_status_real`
  - `test_12_index_base_consistency_checker`
  - `test_13_forecast_real_mode_all_india`
  - `test_14_forecast_real_mode_mumbai`
  - `test_15_forecast_no_synthetic_fallback_in_real_mode`
  - `test_16_unit_preserved_as_index_points`
  - `test_17_geography_level_preserved_as_city`

Express API Tests (`server`)

```
bash
npm test
```

- **Total Tests**: 20 tests (17 baseline + 3 Milestone 4 tests)
- **Passed**: **20 / 20 (100%)**
- **Execution Time**: 43ms
- **New Tests**:
  - `✔ 18. GET /api/forecasting/predict?mode=real&geography=All%20India → 200 status:ok data_status:real`
  - `✔ 19. GET /api/data/sources → RBI-HPI and OSM status is "integrated"`
  - `✔ 20. GET /api/data/series?geography=Mumbai&metric=hpi_assessment → returns real observations`

Client Production Build (`client`)

```
bash
npm run build
```

- **Result**: Passed cleanly in 2.03s (`tsc -b && vite build`).
- **Bundle**: `dist/index.html` (1.01 kB), `assets/index.js` (912.99 kB).

8. Documentation Delivered

1. `docs/REAL_DATA_GUIDE.md` : Step-by-step acquisition, normalization, and deduplication guide for RBI DBIE and OSM amenities.
2. `docs/DATA_DICTIONARY.md` : Complete column definitions, data types, metric classifications, and governance rules.
3. `docs/FORECASTING_EVALUATION.md` : Hold-out split methodology, empirical MAE/RMSE benchmark tables, and documented limitations.
4. `server/python/data/raw/rbi/README.md` : Provenance documentation for raw and normalized RBI files.
5. `README.md` : Updated with dedicated Milestone 4 architecture and capabilities section.

9. Known Boundaries & Next Steps

Known Boundaries

- **City-Level vs. Ward-Level**: Indian official HPI indices exist exclusively at the city level. They cannot and must not be used directly for ward-level gentrification or neighborhood price modeling.
- **Index vs. Price**: Real-mode forecasts yield index points only, never ₹/sq.ft values.
- **NHB RESIDEX**: Manual portal export required due to portal authentication restrictions.

Recommended Milestone 5 Directions

1. **Neighborhood Transformation & Gentrification Risk Scoring**: Build exploratory composite indicators linking real OSM amenity density with simulated micro-market metrics.

2. **Interactive Geospatial Polygon Layer**: Add GeoJSON administrative ward boundaries for Mumbai (Ward A, D, K-West) on the Leaflet map.
3. **Housing Affordability Module Expansion**: Connect household income percentiles with real-world interest rates and loan underwriting guidelines.